SRAM and Flash memory:

- **SRAM (Static Random Access Memory):**
  - Volatile → loses data when power is off.
  - Very fast access time.
  - Used for cache and temporary storage in processors.
  - Requires continuous power to retain data.
  - More expensive and consumes more power compared to other RAM types.
- **Flash Memory:**
  - Non-volatile → retains data even without power.
  - Slower than SRAM for read/write operations.
  - Used for permanent storage (USB drives, SSDs, memory cards).
  - Can be electrically erased and reprogrammed.
  - Cheaper and denser, but with limited write cycles.

Where is DDR register located for each port? (Page 64 at datasheet)

- **DDRA** → Data Direction Register for Port A

Port A Data Direction Register – DDRA

Bit	7	6	5	4	3	2	1	0	
	DDA7	DDA6	DDA5	DDA4	DDA3	DDA2	DDA1	DDA0	DDRA
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **DDRB** → Data Direction Register for Port B

Port B Data Direction Register – DDRB

Bit	7	6	5	4	3	2	1	0	
	DDB7	DDB6	DDB5	DDB4	DDB3	DDB2	DDB1	DDB0	DDRB
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **DDRC** → Data Direction Register for Port C

Port C Data Direction Register – DDRC

Bit	7	6	5	4	3	2	1	0	
	DDC7	DDC6	DDC5	DDC4	DDC3	DDC2	DDC1	DDC0	DDRC
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	

- **DDRD** → Data Direction Register for Port D

Port D Data Direction Register – DDRD

Bit	7	6	5	4	3	2	1	0	
	DDD7	DDD6	DDD5	DDD4	DDD3	DDD2	DDD1	DDD0	DDRD
Read/Write	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W	
Initial Value	0	0	0	0	0	0	0	0	